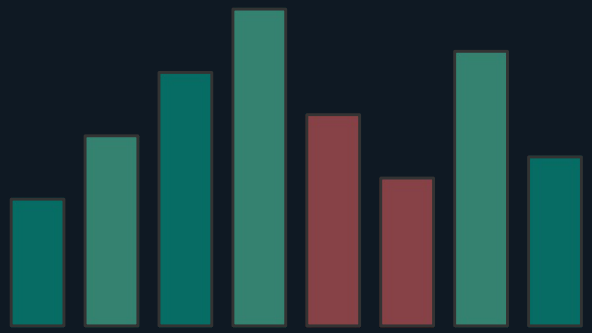
BodySync

Detecting When Your Body's Systems Fall Out of Alignment

Using the Empatica Embrace Plus Wearable
& the ZU-PWD '25 Dataset



The Problem

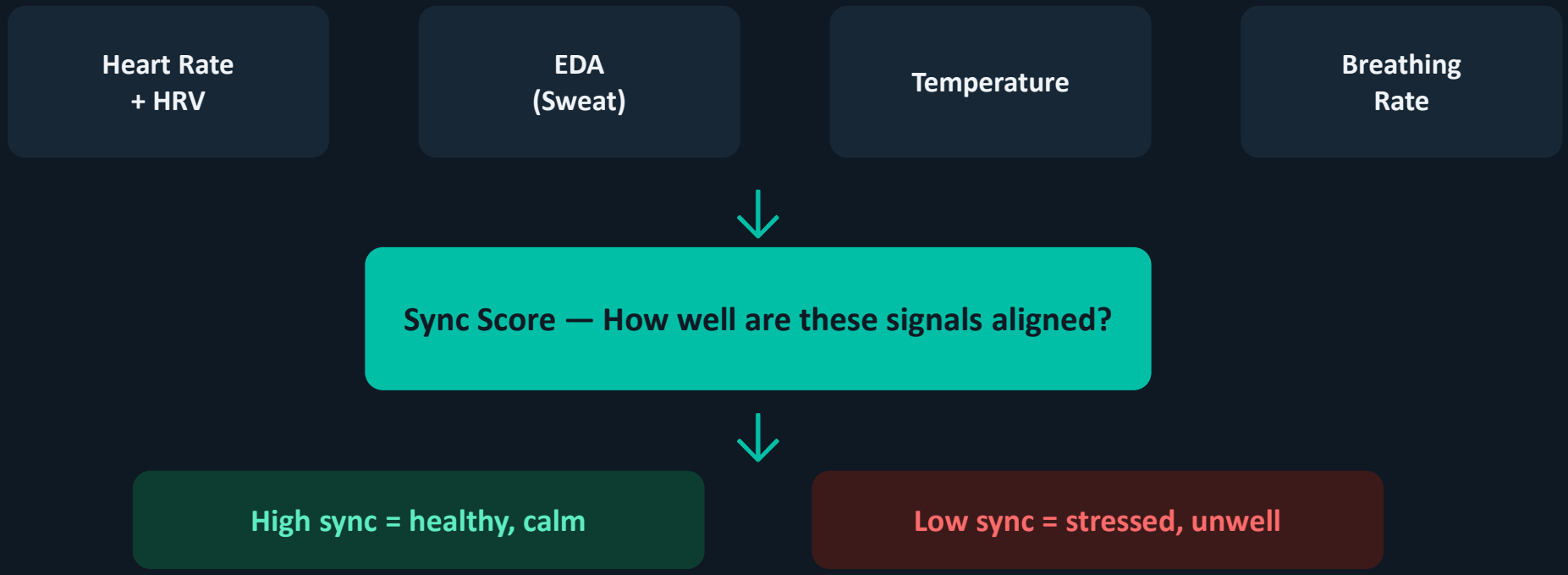
Our bodies send signals constantly — heart rate, sweat, temperature, breathing.
Today's health apps look at each one alone. But the real story is in how they work together.

What the Paper Found (Sun et al., 2026 — IJHCS)

- EDA, HRV, and skin temperature change when people feel trust or distrust toward health info
- They predicted trust levels with 73% accuracy using these signals
- LLM-generated health content was trusted more than human-written content
- Labels ("AI" vs "Human") mattered even more than the actual source
- **Limitation: single time-point, lab setting, signals treated independently**

What Is BodySync?

Instead of looking at one signal at a time, we measure how well all your body's signals agree with each other. When they stop agreeing — that's a "desync."



The Dataset

ZU-PWD ’25 — Zayed University Physiological Wellness Dataset

28	15 mo	901K	24
Participants	Duration	Data Rows	Variables

Signals from the Embrace Plus

- **Pulse Rate**
- **HRV (RMSSD)**
- **EDA (Sweat)**
- **Body Temp**
- **Respiratory Rate**
- **Steps & Activity**
- **Sleep Stages**

Real-world free-living data • Uses Embrace Plus (not the older E4) • Longitudinal — months, not hours

How This Extends the Paper

The Paper Did This	We Extend It To
Measured EDA, HRV, temp separately in a controlled lab	**Measure how signals relate to each other in real daily life**
Single time-point study (one lab session per person)	**15 months of continuous longitudinal data**
Predicted trust with 73% accuracy from individual signals	**Inter-signal coherence (sync score) as a richer combined metric**
Called for ecological assessment and real-world validation	**Free-living data from 28 people in their normal routines**

What We'll Do

1

Explore & Clean the Data

Load the dataset, handle missing values and wear-time gaps, visualize daily patterns for each participant. Understand what the signals look like in real life.

2

Build the Sync Score

Compute rolling correlations between signal pairs (HRV vs temp, heart rate vs EDA, etc). Create per-person baselines. Validate that the score changes between sleep, rest, and activity.

3

Predict Desyncs with ML

Label episodes where the sync score drops. Train Random Forest and XGBoost models to predict upcoming desyncs. Use SHAP to explain which signals matter most.

4

Visualize & Report

Build "body weather maps" using UMAP. Show how each person's physiological state moves over days and weeks. Connect findings back to the paper's trust framework.

Tools & Methods

`Python` `pandas` `scikit-learn` `XGBoost` `SHAP`

`matplotlib` `seaborn` `UMAP` `scipy`

Sync Score

Rolling Pearson correlation between signal pairs over sliding windows (1h, 6h, 24h). Averaged into one coherence metric per window.

Personal Baselines

Per-person, per-activity-context normal ranges. Deviations beyond 2 std devs flagged as desync events.

Classification

Random Forest and XGBoost to predict upcoming desync. Leave-one-subject-out cross-validation. SHAP for explainability.

Body Weather Maps

UMAP dimensionality reduction to project all signals into 2D. Track how each person moves through the state space over time.

Where This Can Go

Validate with labeled data	Test the sync score on WESAD or other datasets that have stress/emotion ground truth
Build a real-time dashboard	Live system that computes sync score from streaming Embrace Plus data
Replicate the trust study	Run the paper’s experiment with participants wearing Embrace Plus, using sync score as the outcome
Adaptive health app	Prototype an app that adjusts health notifications based on detected sync state
Clinical deployment	Partner with health researchers to test if desync predicts illness in patients